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Reduction of the fault current passing through the grounding system of an HV substation supplied by cable line

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ABSTRACT

The paper presents an original analytical procedure developed for the determination of the part of ground fault current dissipated through the substation grounding system (electrode) into the surrounding earth in the cases when a specific measure has been taken to reduce it. The practical realization of this measure is simple and consists of burying one bare copper wire of a sufficient length together with the feeding cable line. The presented method enables a quick, and, for practical purposes reasonably accurate determination of the copper wire length providing the required safety conditions in the considered HV distribution substation and its vicinity. The developed analytical procedure is simplified by introducing an equivalent neutral conductor substituting all three cable line metal sheaths as well as all surrounding metal installations. The presented quantitative analysis shows that the effects achieved by this measure are significant.

1. Introduction

During a phase to ground fault in an HV (high voltage) electric-power network with directly earthed neutral points, increased potentials appear at places where they normally do not exist. In order to ensure protection against undesired consequences, such as the loss of human lives and damages to sensitive electronic and telecommunication equipment using in the substation, it is necessary to evaluate the value of these potentials as accurately as possible [1–3] and put them in the prescribed limits by designing an adequate grounding system.

The magnitudes of the raised potentials as well as of all the potential differences significant for the safety conditions (touch and step voltages) are proportional to the ground fault current portion dissipating through the substation grounding system into the surrounding earth. The remaining parts of the ground fault current return to their sources in power system through different metal paths (cable line metal sheaths and surrounding metal installations typical for urban areas) and earth connections, so that they do not contribute to the creation of the substation grounding electrode potential [1,2,5–7]. Thus, only after correct determining this ground fault current portion, it is possible to put the whole problem into a realistic framework and to arrive at safe solutions without excessive expenditures. Therefore, when designing substation grounding system it is of prime importance to determine, as accurately as possible, ground fault current distribution.

In the case of cable feeding lines that can be treated as homogeneous along their entire length, the analytical expressions for determination ground fault current distribution are developed in [1–3]. However, by using them it is not possible to take into consideration the existence of the metal installations surrounding a feeding cable line. The method enabling appreciation of the favorable inductive influence of surrounding metal installations, i.e. determination of the actual feeding line reduction factor in each concrete case was developed in [8–11]. As shown, surrounding metal installations typical of urban areas can significantly reduce the portion of the ground fault current dissipated through the substation grounding system. This fact dramatically has changed our former general impression concerning the magnitude of the grounding problem of HV substations located in urban areas.

Also, our perception of grounding problem of these substations is nowadays more favorable because of our knowledge of the existence of a global earthing system around many of these substations [12,13]. However, in some cases, relevant conditions and circumstances can be very unfavorable. They are encountered in practice as different combinations of a high value of soil resistivity, high level of the ground-fault current, a small number of metal installations surrounding feeding line, a small number of external grounding electrodes, confined yard foreseen for substation, substation surrounded by relatively old buildings (without foundation earth electrodes), etc. An especially unfavorable circumstance is the fact that at present in the MV (middle voltage) distribution networks different type cables having insulated outer sheath are replacing, more and more, the paper-insulated cables whose grounding contributions have been long time considered as essential for safe and cost-effective solving of the grounding problems of HV distribution substations. Under such practical conditions and

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circumstances, the calculated value of the substation grounding electrode potential can be significantly higher than one kilovolt and its reduction becomes a mandatory demand in front of the substation designers. Then, a reduction of the portion of the ground fault current dissipated into the surrounding earth can be the only solution. It can be achieved by ordering a nonstandard HV cable construction that has a sufficiently small value of the reduction factor. However, especially if the feeding cable line is relatively long, this solution may prove itself very expensive. Thus, another way of solving this problem should be considered.

The more favorable ground fault current distribution can be also achieved by laying a bare copper wire in the same trench and on the certain length along the feeding cable line [4]. As it will be shown in Section 5 this way of solving an HV substation grounding problem is more cost effective even under an unreal assumption that a sufficiently large grounding grid can be laid in the certain urban area. There are at least two impotent reasons to that. First of all, a strong inductive coupling of the buried copper wire with the cable line conductors intensifies the fault current dissipation into the ground through this wire. Secondly, there is no need to dig a separate trench to lay down this wire. Finally, this solution opens the possibility of applying the HV cables of standard construction, even in extraordinary unfavorable grounding conditions.

The first analytical method developed in the aim of evaluation of the effects achieved by the considered measure was based on the too rough approximation that the current through the buried copper wire has no any inductive influence on the value of the currents through the cable line metal sheaths [4]. This paper introduces in it an improvement which enables taking into consideration the fact that the current circulating through the buried copper wire has a certain inductive influence on the currents through the cable line metal sheaths. Also, on the bases of the introducing an equivalent neutral conductor (the research achievement presented in [8–11]), the presented method enables taking into the consideration the very favorable inductive influence of metal installations surrounding each HV distribution cable line.

2. Problem description

The mentioned measure, by providing an additional return path for the ground fault current, leads to an overall reduction of potentials and voltages appearing in the supplied HV distribution substation and in its vicinity. However, the problem is: how one can determine the achieved degree of this reduction? By installing the copper wire on certain length along the feeding cable line, this line cannot be anymore considered as homogeneous and the existing methods for determination ground fault current distribution cannot be treated as adequate. Also, since the considered problem appears in the cases that are critical from standpoint of safety conditions, it is necessary to take into consideration the favorable inductive influence of the metal installations surrounding feeding lines. Thus, the whole problem becomes very complex.

As an illustration, the considered measure for obtaining a more favorable ground fault current distribution is symbolically presented in Fig. 1.

The used notation has the following meaning:

A (B) – supply (supplied) substation.

On the basis of Fig. 1, it is clear that the buried copper wire has a complex function, i.e. at the same time, it can be considered as an external electrode of the grounding system of substation B and a specific neutral conductor of the considered feeding cable line. Thus, the problem that should be solved can be defined as: how can one determine the portion of the ground fault current absorbed at substation B by the buried copper wire? This portion of the ground fault current is diverted from the substation grounding electrode and at the point of discontinuity D (end of the buried copper wire) continues its flow toward the grounded neutral points in the HV network through the earth. It appears as a consequence of the conductive coupling between the copper wire and the substation grounding electrode (usually constituted only of substation building foundation), as well as a consequence of the inductive coupling between this wire on one side and cable line conductors and surrounding metal installations typical for urban areas, on the other side. Thus, under practical conditions, the magnitude of this current is a very complex function of the self and mutual impedances of the: copper wire, cable line conductors and surrounding metal installations, substation grounding impedance, local soil resistivity as well as many other factors of a lower order.

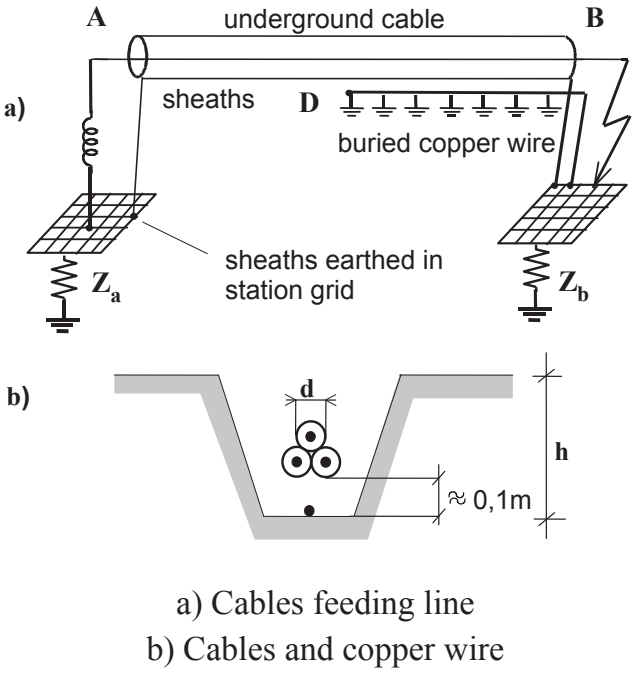


Fig. 1. A non-homogeneous cable line.

3. Equivalent cable line metal sheath

Let us assume that the HV feeding line presented in Fig. 1 is consisting of three single-core cables laid in flat formation with spacing, the general case from the standpoint of the mutual disposition of single-core cables, Fig. 2. Also, let us assume at the beginning that there is no any metal installation in its vicinity.

With the aim to determine only the inductive influence of the cable line metal sheaths on the ground fault current distribution we will make the following assumptions:

- the grounding impedances of the substations at the ends of the line are negligible, $Z_a \approx 0$ and $Z_b \approx 0$, and
- the currents in the cable line sheaths and the buried copper wire have a negligible inductive influence on the value of total ground fault current, I_f.

Then, the corresponding equivalent circuit of the considered HV

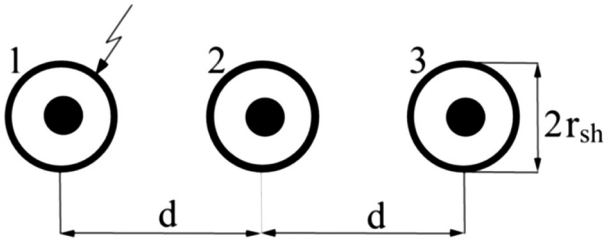


Fig. 2. Single-core cable flat formation with spacing.

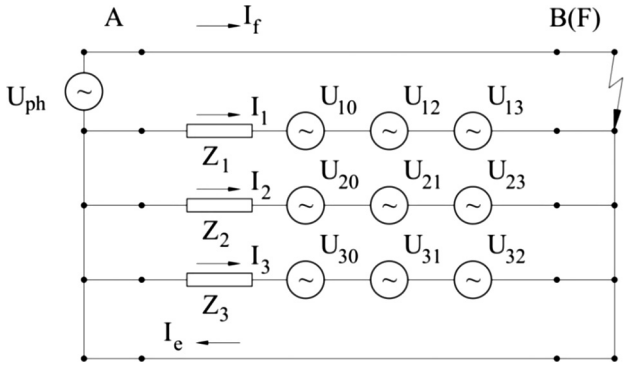


Fig. 3. Equivalent circuit of the adopted cable line.

cable line during the ground fault in the substation B is presented in Fig. 3.

The used notation has the following meaning:

U_{ph} – phase voltage in the supply substation, A,
U_{10}, U_{12}, U_{13}, – voltages induced in the metal sheath of the cable with ground fault current I_f,
U_{20}, U_{21}, U_{23}, U_{30}, U_{31}, U_{32} – voltages induced in two other metal sheaths,
I_f – ground fault current through the faulted phase conductor of the considered cable line,
I_1 – current through the sheath of the cable with the current I_f,
I_2, I_3 – currents through the metal sheaths of the other two single-core cables, and
Z_1, Z_2, Z_3 – self- impedances of cable metal sheaths ($Z_1 = Z_2 = Z_3 = Z_{sh}$).

For the sake of simplifying the necessary calculation procedure, we will introduce an equivalent cable line metal sheath substituting all three actually existing metal sheaths, Fig. 2.

Under such assumption the considered cable line is, analogously to the procedure described in [8–11], transformed into the simpler physical model whose cross-section is represented in Fig. 4.

For the imagined physical appearance of the equivalent conductor (sheath) presented in Fig. 4, the corresponding equivalent circuit of the entire cable line can be presented as shown in Fig. 5.

The relevant parameters of the equivalent sheath can be determined under the condition that current I_e in the equivalent circuit in Fig. 3

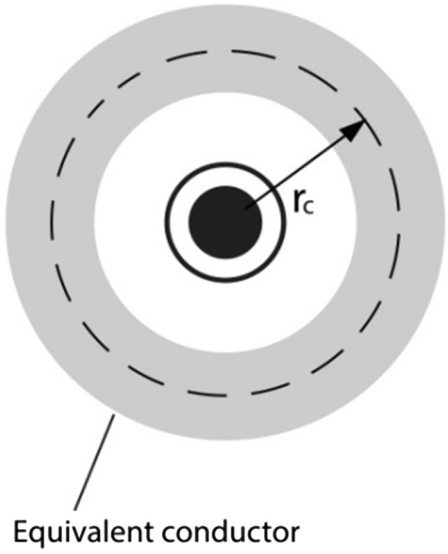


Fig. 4. Cross section of the faulted phase conductor and equivalent metal sheath.

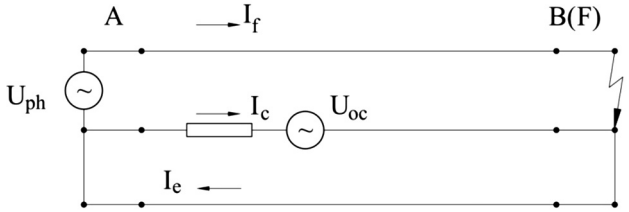


Fig. 5. The equivalent circuit of the line model in Fig. 4.

remains unchanged after its introducing, Figs. 4 and 5. According to the definition of the feeding line reduction factor [2], this means that the reduction factor of the actual cable line (Fig. 2) remains unchanged after the introducing the equivalent conductor. Thus, this condition can be, according to Figs. 4 and 5, expressed by

$$r_e = \frac{I_e}{I_f} = \frac{R'_C}{R'_C + \omega\frac{\mu_0}{8} + j\frac{\omega\mu_0}{2\pi}\left(\frac{\mu_r}{4} + \ln\frac{\delta}{r_c}\right)} = r_d \tag{1}$$

where

r_e – reduction factor of the cable line model presented in Fig. 4,
r_d – previously determined value of the reduction factor of the feeding cable line,
r_c – medium radius of the cylinder representing the equivalent sheath, and
R'_c – the equivalent sheath resistance, per unit length, Ω/km.
ω – angular frequency $\omega = 2\pi f$,
μ_0 – magnetic permeability of vacuum: $4\pi\ 10^{-7}$ (Vs/Am), and
δ – equivalent earth penetration depth.

The equivalent earth penetration depth is determined by

$$\delta = 658\sqrt{\frac{\rho}{f_n}},\ \text{m}, \tag{2}$$

where

ρ – equivalent soil resistivity along and around the cable line (Ωm), and
f_n – nominal power frequency, 50 or 60 Hz.

The reduction factor of the cable lines consisting of three single-core cables in the formation shown in Fig. 2 is, according to [2], determined by

$$r_d = \frac{R'_{sh}}{Z'_{sh} + Z'_{12} + Z'_{13} - Z'_{12}\frac{Z'_{12} - Z'_{13}}{Z'_{sh} - Z'_{12}}}. \tag{3}$$

where

R'_{sh} – resistance of cable sheath per unit length,
Z'_{sh} – self-impedance of cable sheath per unit length,
Z_{12} – mutual impedances between cable 1and cable 2 in Fig. 2,
Z_{13} – mutual impedances between cables 1 and 3 in Fig. 2.

The self-impedance of the cable sheath per unit length is, according to [3] determined by

$$Z'_{sh} = R'_{sh} + \omega\frac{\mu_0}{8} + j\omega\frac{\mu_0}{2\pi}\left(\frac{\mu_r}{4} + \ln\frac{\delta}{r_{sh}}\right),\ \Omega/\text{km} \tag{4}$$

while the mutual impedance between two adjacent cables, denoted by Z'_{12}, is determined, according to [3], by the following expression

$$Z'_m = \omega\frac{\mu_0}{8} + j\frac{\mu_0}{2\pi}\ln\frac{\delta}{d},\ \Omega/\text{km} \tag{5}$$

where

r_{sh} – medium radius of the cable metal sheath, and
d – the separation between two nearby cables (Fig. 2).

Mutual impedance between the two outer cables (Fig. 2), denoted by Z'_{13}, is also determined by expression (5) only, according to Fig. 2, "d" is replaced by "$2d$".

In the most frequent case in practice, when single-core cables are laid in the triangular formation (Fig. 1b) instead of (3), we have, according to [2], the following expression

$$r_d = \frac{R'_{sh}}{R'_{sh} + 3\omega\frac{\mu_0}{8} + j\omega\frac{\mu_0}{2\pi}\left(\ln\frac{\delta}{r_{sh}} + 2\ln\frac{\delta}{d}\right)}. \quad (6)$$

Since (1) gives the relationship between complex quantities, the condition for obtaining the equivalent cable line sheath can be expressed by the following system of two equations

$$\begin{aligned} &\mathrm{Re}\left(\frac{R'_C}{R'_C + \omega\frac{\mu_0}{8} + j\omega\frac{\mu_0}{2\pi}\left(\frac{\mu_0}{4} + \ln\frac{\delta}{r_c}\right)}\right) = \mathrm{Re}(r_d) \\ &\mathrm{Im}\left(\frac{R'_C}{R'_C + \omega\frac{\mu_0}{8} + j\omega\frac{\mu_0}{2\pi}\left(\frac{\mu_0}{4} + \ln\frac{\delta}{r_c}\right)}\right) = \mathrm{Im}(r_d). \end{aligned} \quad (7)$$

where

Re(r_d) –real part of the cable line reduction factor determined by (3) or (6), and
Im(r_d) – imaginary part of the cable line reduction factor determined by (3) or (6).

However, the actual reduction factor depends on many local factors and cannot be determined only by calculations, i.e. by using (3) or (6). It should be separately determined in each concrete case by using the method developed on the basis of the result of the-on-site measurements [8–11]. The equivalent neutral conductor parameters (R'_C and r_c) determined on the basis of such reduction factor cumulatively involve all relevant factors and parameters including the very favorable inductive influence of all, known and unknown, surrounding metal installations typical for urban areas.

4. The simple equivalent circuit

Let us assume that substation A is either source or transit one and that substation B is supplied by a cable line, as shown in Fig. 1. Also, let us assume that both of these substations belong to an HV network with directly grounded neutral points and that the current appearing in the foreseen copper wire has a negligible influence upon the value of the total ground fault current, that is certainly a realistic assumption.

After determining the relevant parameters of the equivalent sheath (R'_C and r_c), by using as the theoretical foundations:

- the method of symmetrical components,
- the decoupling technique, and
- the reduction of uniform ladder circuits to the one equivalent π-network [4],

the grounding system is shown in Fig. 1 can be presented by the relatively simple equivalent circuit in Fig. 6.

The used notation has the following meaning:

U_1 – voltage induced by current I_f in the equivalent cable line sheath,
U_2 – voltage induced by current I_i in the equivalent cable line sheath,
U_3 – voltage induced by current I_f in the buried copper wire,

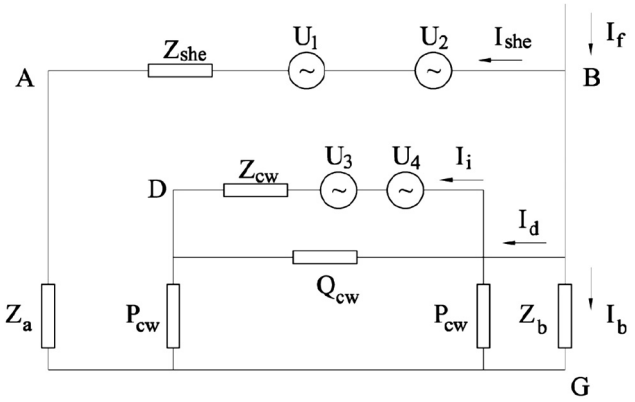


Fig. 6. The equivalent circuit of the grounding system of Fig. 1.

U_4 – voltage induced by current I_{she} in the buried copper wire,
I_f – ground fault current component passing through the grounded neutral points in the supply network,
I_{she} – fault current through the equivalent cable line sheath,
I_b –fault current dissipating into the earth through the substation grounding system excluding the buried copper wire,
I_d – fault current diverted from the substation grounding system by the buried copper wire,
I_i – current resulting from the influence of the inductive coupling between the buried copper wire on one side and the cable line conductors and surrounding metal installations on the other side,
Z_{she} – self-impedance of the equivalent cable line sheath,
Z_{cw} – self-impedance of the buried copper wire,
Q_{cw} and P_{cw} – impedances of the equivalent π-network for the lumped parameter ladder circuit substituting the buried copper wire as long grounding conductor,
Z_a (Z_b) – grounding system impedance of substation A (B), and
G – remote ground.

For the known relevant parameters of the buried copper wire including its length and the equivalent resistivity of the surrounding soil, impedances: Z_{cw} can be, according to e.g. [3], determined by

$$Z_{cw} = Z'_{cw}L = \left[R'_{cw} + \omega\frac{\mu_0}{8} + j\omega\frac{\mu_0}{2\pi}\left(\frac{\mu_r}{4} + \ln\frac{\delta}{r_{cw}}\right)\right]L,\ \Omega \quad (8)$$

where

R'_{cw} – copper wire resistance (Ω/km),
r_{cw} – radius of the copper wire (m),
ω – angular frequency, $\omega = 2\pi f$,
μ_r – relative permeability of copper wire,
μ_0 – magnetic permeability of vacuum: $4\pi\ 10^{-7}$ Vs/Am,
δ – equivalent earth penetration depth (m), and
L – buried copper wire length (km).

The equivalent impedances Q_{cw} and P_{cw} are, according to [4], determined by

$$Q_{cw} = \frac{k^{L'} - k^{-L'}}{2} Z_{cw\infty} \quad (9)$$

$$P_{cw} = \frac{k^{L'} + 1}{k^{L'} - 1} Z_{cw\infty}, \quad (10)$$

where

L' – relative copper wire length expressed in relation to the length of one meter, and
$Z_{cw\infty}$ – input (grounding) impedance of the infinitely prolonged

copper wire.

The parameter k represents the "current distribution factor" in the nodes 1, 2, 3…, of the discrete parameter ladder circuit representing the buried copper wire under the assumption that it is infinitely prolonged ($L \rightarrow \infty$). This parameter is determined by

$$k = 1 + \frac{Z_{cw\infty}}{R_S}, \tag{11}$$

where

$$Z_{cw\infty} = \frac{Z_S}{2} + \sqrt{R_S Z_S + \frac{Z_S^2}{4}}. \tag{12}$$

Since the adopted section of discretization is, for practical reasons, only one meter ($R_S \gg |Z_S|$) instead of (12), for the determination of impedance $Z_{cw\infty}$, one can use the following approximation

$$Z_{cw\infty} \approx \sqrt{R_S Z_S}. \tag{13}$$

Impedances Z_S is the self-impedance of the cable sheath on the length of one meter, or $Z_S = Z'_{sh} \times 1\,\text{m}$.

Under the assumption that the cable sheath is an equipotential electrode over its entire length, the grounding resistance of the metal sheath on the cable line section of one meter, according to [14], is

$$R_S = \frac{\rho}{\pi\ m} \ln \frac{L}{\sqrt{h d_{cw}}},\ \Omega \tag{14}$$

where

ρ – equivalent soil resistivity along and around the cable line (Ωm),
L – total copper wire length (m),
h – depth in which the copper wire is laid (m), and
d_{cw} – outer diameter of the copper wire (m),
m – note for one meter of length.

Since cable lines are used almost without any exception in the urban areas the value of equivalent soil resistivity ρ of the surrounding area-earth, cannot be exactly determined. Although there are several methods of measuring soil resistivity, none of them is accurate in urban conditions, e.g. [2]. The objective obstruction represents the fact that surfaces of urban areas are covered by buildings, streets, pavements, and many other permanently constructed objects. Also, many known and unknown metal installations have already existed under the ground surface and affect measurement results.

Because of that equivalent soil resistivity ρ can be only approximately estimated on the basis of the main geological characteristics of the soil in which the cable line is laid. By this it is important to know the following: former investigations of soil effects on the grounding impedance of long grounding conductors showed that the adoption of an equivalent soil resistivity somewhat higher (up to 20%) than the one approximately estimated ensures final results slightly on the safe side [4]. Thus an accurate evaluation and final judgment about the achieved safety conditions can be obtained only on the basis of measurements that are, as it is well known, mandatory before putting a newly built substation in operation.

On the basis of previously determined R'_C and r_c, for the certain cable line length it is possible to determine impedance Z_{she} and all induced voltages: U_1, U_2, U_3 and U_4. Since impedances: Z_a, Z_b, and Z_{she} also can be determined, by using the equivalent circuit in Fig. 6 one can determine current I_b necessary for estimation of the achieved safety conditions.

However, for determination of the reduction factor of this specific feeding line it is necessary to assume that impedances Z_a and Z_b are negligible ($Z_a \approx 0$ and $Z_b \approx 0$). Also, the significant effects of the considered measure are obtained when the length of the copper wire is so large that impedance Q_{cw} can be considered as infinitely large. Thus, for the sake of simplicity we will adopt that this impedance is infinitely large ($Q_{cw} \rightarrow \infty$) giving the final result slightly on the safe side. By simplifying the equivalent circuit (Fig. 6) on the basis of the previous approximations, we can write the following system of equations

$$Z_{she} I_{she} + Z_{m1} I_f + Z_{m2} I_i = 0 (Z_{cw} + P_{cw}) I_i + Z_{m3} I_f + Z_{m4} I_{she} = 0 \tag{15}$$

where

Z_{m1} – mutual impedance between the faulted phase conductor and the equivalent cable line sheath,
Z_{m2} – mutual impedance between the equivalent cable line sheath and the buried copper wire,
Z_{m3} – mutual impedance between the faulted phase conductor and the buried copper wire, and
Z_{m4} – the mutual impedance between the equivalent cable line sheath and the buried copper wire.

Bearing in mind the imagined physical appearance and disposition of the equivalent conductor, Fig. 4, one can see that the impedances: Z_{m2}, Z_{m3}, and Z_{m4} are mutual equal ($Z_{m2} = Z_{m3} = Z_{m4}$) and can be commonly denoted as Z_m. On the basis of this observation and by solving system of Eq. (15) we obtain

$$I_{she} = \frac{Z_m^2 - (Z_{cw} + P_{cw}) Z_{m1}}{(Z_{cw} + P_{cw}) Z_{she} - Z_m^2} I_f \tag{16}$$

$$I_i \approx \frac{Z_m Z_{m1} - Z_{she} Z_m}{(Z_{cw} + P_{cw}) Z_{she} - Z_m^2} I_f. \tag{17}$$

On the basis of (16) and (17), the reduction factor of the considered feeding line (Fig. 1) is

$$r_i = \frac{I_f - I_{she} - I_i}{I_f} \approx \frac{(Z_{cw} + P_{cw} - Z_m)(Z_{she} - Z_{m1})}{(Z_{cw} + P_{cw}) Z_{she} - Z_m^2}. \tag{18}$$

Finally, on the basis of (18) the reduction of the feeding line reduction factor obtained only by the buried copper wire, r_{cwi}, is

$$r_{cwi} = \frac{r_i}{r_d} \approx \frac{(Z_{cw} + P_{cw} - Z_m)(Z_{she} - Z_{m1})}{[(Z_{cw} + P_{cw}) Z_{she} + Z_m^2] r_d}. \tag{19}$$

Then, on the basis of the approximation that the impedance Q_{cw} is infinitely large, the reduction of current I_b resulting only as a consequence of the conductive couplings between the cable line sheaths and copper wire on one side and the substation grounding electrode on the other side, according to Fig. 6, is

$$r_{ct} = \frac{P_{cw} Z_{she}}{(P_{cw} + Z_{she}) Z_b + P_{cw} Z_{she}}, \tag{20}$$

while only as a consequence of the conductive coupling between the copper wire and the substation grounding electrode, is

$$r_{cwc} = \frac{P_{cw}}{Z_b + P_{cw}}. \tag{21}$$

Finally, the current passing through the grounding system of supplied substation B is determined by

$$I_b = r_t I_f = r_i r_{ct} I_f, \tag{22}$$

where r_t represents the total relative reduction of current, I_b.

Also, the total effect of the copper wire on the reduction of this current can be determined by

$$r_{cwt} = r_{cwi} r_{cwc}. \tag{23}$$

In the case when the copper wire is laid along the entire cable line length the given expressions should be modified. Thus, in (18) and (19) instead, P_{cw} should be Z_a if substation A is transit one or should be omitted ($P_{cw} = 0$) if substation A is source one. Also, if the copper wire is long nearly its active length [14] and longer in (20) and (21) instead P_{cw} should be impedance $Z_{cw\infty}$.

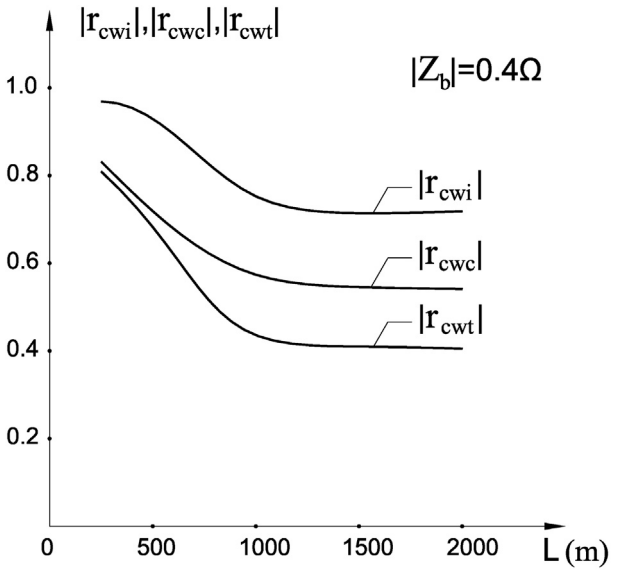


Fig. 7. Coefficients: r_{cwi}, r_{cwc} and r_{cwt} owing to the copper wire length.

Since dimensions of grounding electrode of the considered type of HV substations are relatively small the proximity effect between the substation grounding electrode and buried copper wire can be disregarded [15].

5. Quantitative analysis

This analysis is performed with the aim to create a general impression about the efficiency of the here presented measure for the reduction of the fault current dissipated through the grounding system (electrode) of a supplied HV distribution substation.

Let us assume a substation that is supplied by 110 kV cable line long three kilometers. The foreseen cable type is XHE 94-A and in the standard design, its sheath has the other diameter equal 91 mm and is made of copper strings with cross-section equal 95 mm^2. Also, we will assume that there is no any metal installation in its vicinity.

The reduction factor of such cable line when its cables are laid in triangular formation, by using (6), is $r_d = 0.020 - j0.106$. Then, by selecting the next larger sheath cross-section offered by the manufacturer's catalog equal 120 mm^2, the value of the cable line reduction factor would be 0.015 − j 0.084. If a higher reduction of current I_b is necessary, the next larger cross-section is 185 mm^2 and the reduction factor of the considered cable line would be: 0.008 − 0 j0.055.

We will consider the effects of the here described measure under the assumption that the buried copper wire has a cross-section of 120 mm^2 and is buried in the depth of one meter together with the line cables, Fig. 1. Also, we will assume that the equivalent soil resistivity is relatively high or $\rho = 200\,\Omega m$ and that the effective value of grounding impedance Z_b is also relatively high, or $Z_b = 0.4\,\Omega$.

On the basis of the given data, by using (20) and (21) the variations of r_{cwi}, r_{cwc}, and r_{cwt} as a function of the copper wire length are graphically presented in Fig. 7.

According to the given diagrams, there is no any purpose of laying the copper wire longer than the certain limiting length. As can be seen in the considered case this length is about one kilometer.

In the cases when the substation grounding electrode (grid) has been already reinforced through the use of vertical earth rods, satellite earth grids, etc, and a farther reduction of impedance Z_b is not possible, then, the designers, generally speaking, have only the following two possibilities:

– application of the measure considered herein, or
– placing an order for the manufacturers to fabricate a specially constructed cable having the sufficiently small value of the reduction factor.

Then, of course, the following question is raised: which of these two solutions is more cost-effective?

On the basis of the given data, it is not difficult to show that the reduction of 24.36% and 50.7% obtained by successive enlarging cable sheath cross-section requires the additional consumptions of the copper strings that are: 1990 and 7168 kg, respectively. However, for the reductions of: 29.3% (L = 500 m), and 66.8% (L = 1000 m) obtained by the buried copper wire it is necessary only: 531 kg, 1062 kg, respectively.

Based solely on the presented results, it is obvious that the significant advantage is on the side of the measure presented herein i.e. it is more economical to invest in the copper wire than in increase the cross-section of the cable line metal sheaths.

Of course, such favorable effects can be explained by the fact that the copper wire acts as an external electrode distant from other elements of the substation grounding system (electrode). Beside of that, the additional fault currents are diverted by this wire because of its inductive coupling with the feeding cable line conductors. All these, certainly, means that the dissipation of the ground fault current through this copper wire is more efficient than through the copper wire within the substation grid. Also, it is important to mention that the effect of reduction is more pronounced when it is more needed, i.e. when the value of grounding system impedance Z_b is higher.

In practical conditions, because of the inductive influence of surrounding metal installations, the participation of the buried copper wire will be smaller, i.e. the value of reduction coefficient r_{cwi} will be somewhat greater, however, the reduction factor of the considered cable line r_i will be smaller, i.e. more favorable [8]. Certainly, the inductive influence of surrounding metal installations is greater if the considered line passes through the area with a higher degree of urbanization.

6. Conclusions

The developed analytical method enables, at the design stage, a relatively easy and correct estimation of the reduction degree of the fault current through an HV distribution substation grounding system achieved by the considered measure. The complete application of the developed method enables us to take into the consideration all relevant factors and parameters including the inductive influence of surrounding metal installations. The reduction of the ground fault portion dissipated through the substation grounding system into the surrounding earth by using the considered measure is remarkable.

The need for the considered measure can be expected mainly in the case of HV distribution substations situated deep within the city area, where normally there is no available space for laying sufficiently large grounding electrode and where there are many relatively old buildings without foundation earth electrodes.

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